



Michael Boiman

Freelance Quality Engineer & AI Architect • 20+ Years QA • 4+ Years AI Development
Frankfurt am Main, Germany · Freelance · available for new projects

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 BKS-Lab Organization

40+

Skills

one orchestration hub

20+

Repos orchestrated

across 7 project boards

Live

A2A agent

live at mboiman.bks-lab.com

Quality Engineering | LLM/AI Architecture | CI/CD Automation | Multi-Cloud | Peppol/E-Invoice

Career Profile

Quality Engineer and AI Automation Expert with 20+ years in software quality assurance and 4 years of specialized AI development. Quality engineering is not a closed chapter but an ongoing present — both tracks feed into each other.

In enterprise projects (DB Vertrieb, DVAG, TÜV Süd), quality monitoring systems took shape, AI-driven test automation was added, and legacy migrations were secured with 24/7 validation.

Today I combine quality engineering with AI architecture: MCP and A2A protocols, multi-agent systems, and e-invoicing to EU standards — bundled into one orchestration hub across 20+ repos, agents, and protocols.

I also pass this knowledge on — in developer and business workshops and as an invited guest speaker at TU Darmstadt — translating hands-on AI for technical and non-technical audiences alike.

Technical Skills

Quality Engineering Playwright

Cucumber/Gauge Python

LLMs (Claude/Gemini/GPT/Copilot)

Agent Orchestration (MCP & A2A)

Google ADK LangChain

Peppol/E-Invoicing (EN 16931)

CI/CD Kubernetes

Azure Functions Grafana

Elasticsearch REST APIs

Docker JMeter Gatling

Education

Diploma in Computer Science (UAS)

Cologne University of Applied Sciences, Gummersbach
2000 – 2005

ISTQB Certified Tester

2005

Certified Dynatrace Diagnostics Basic Training

2007

Languages

German Native
English Professional

Additional Qualifications

ISTQB Certified

Agile (Scrum/Kanban)

4+ Years AI Development, 20+ Years Quality Engineering

PROJECTS & SOLUTIONS

Quality Dashboard - Real-time Overview

Grafana Elasticsearch Python CI/CD Playwright

End-to-end solution for real-time quality monitoring across 10 test tools, 5 environments plus multi-source integration (Pipeline, LeanIX, Jira, Confluence) for automated PDF reports with open issues, bugs, and technical debt.



24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java Cucumber Kibana Graylog Elasticsearch

Enterprise Legacy Transformation with End-to-End Quality Assurance

The Challenge: Complex mainframe price calculation with millions of combination possibilities needed secure migration - without quality loss or data errors. **The Solution:** 24/7 Dual-Run Validation System with intelligent deviation analysis • **Continuous Testing:** Randomly generated tests across all variations around the clock • **Real-time Monitoring:** Live dashboard with immediate deviation detection on...



E-Invoicing Automation Platform

Python Azure Functions Peppol Storecove SAP ByDesign

Production E-Invoicing Pipeline (Peppol/EN 16931) Core Features: • **Peppol Network Integration:** Connection to European e-invoicing network via Storecove Access Point • **Multi-Format Support:** ZUGFeRD, Factur-X, XRechnung, UBL 2.1, CII (EN 16931 compliant) • **ERP Integration:** Automatic transfer to SAP ByDesign with validation and error handling • **Hexagonal Architecture:** Domain-driven design with Ports & Adapters for maximum testability • **Dual-Channel Inbound:**...

AI Context Orchestrator — Multi-Repo Development Platform

Python YAML Claude Code Plugins Hooks API GitHub Actions

Central Control System for Enterprise AI Development Core Features: • **Context Orchestrator:** ecosystem.yaml as single source of truth for 20+ repos, customers, infrastructure and skills • **40+ Skills:** Log analysis, deployment verification, incident handling, email management, meeting transcription, PDF/certificate generation, SharePoint integration • **Hook-based Permission System:** Smart Guard with project-specific security rules (branch protection, secret detection, deployment gates) • ****Structured...**

Email Classification and Processing Workflow

Python Azure Functions OpenAI GPT-3.5 Microsoft Graph API Elasticsearch

Complete automation solution with Microsoft Graph API, SAP RFC, Azure Functions (Python), OpenAI GPT-3.5-Turbo, Elasticsearch 8.x, Kibana Dashboards, Docker containers, and GitHub Actions CI/CD.



Developer Workshop: AI-Agent Development & Integration

Anthropic Claude Google Gemini OpenAI GPT-4o MCP A2A

Target Audience: Developers, Technical Leads, DevOps Engineers **Workshop Content:** • **Hands-on AI Agent Development:** Live coding with Claude, Gemini, GPT-4o • **Next-Gen Protocols:** MCP & A2A implementation for enterprise integration • **Browser Automation:** Playwright-MCP for automated test generation • **IDE Integration:** AI assistance in VS Code, Cursor, GitHub Copilot workflows • **Multi-Agent Orchestration:** JSON-RPC, secure agent communication • **Practical Demos:...**



Professional Experience

LLM Infrastructure Architect & Automation Engineer

06/2025 – present

BKS - AI Research & Development

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

Key Responsibilities

- **Open Source MCP Server Portfolio:** Development and publication of 3+ MCP servers on GitHub (bks-wiki-mcp, hubspot-mcp-bks, bks-codex) with uvx distribution, Claude Desktop integration, and multi-format support (Markdown, JSON, YAML)
- **Multi-Agent Orchestration System:** Implementation of production-ready A2A system with Orchestrator Agent (routing hub) and Knowledge Agent for inter-agent communication, capability-based Agent Cards, and JSON-RPC 2.0
- **Git Wiki Transformation:** Migration of Confluence knowledge base to structured Git-based wiki with hierarchical organization and multi-repository architecture (submodules for customer projects)
- **MCP Navigation Server:** Development of Model Context Protocol server (Python) with intelligent hierarchy navigation, automatic content discovery, and SharePoint integration
- **Google ADK Integration:** Implementation of Google Agent Development Kit for multi-agent systems and advanced tool integration

Tools & Technologies

- LLM Orchestration: Claude, Google Gemini, OpenAI GPT, GitHub Copilot, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration
- Multi-Agent: A2A Protocol, Orchestrator/Knowledge Agents, Agent Cards, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Clockify API, Elasticsearch, Natural Language Processing
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems; orchestrated repos span Python, TypeScript, Rust and Go

Quality Engineering

09/2025 - present

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Key Responsibilities

- **BDD API Test Framework:** Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- **E2E Test Automation:** Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- **Performance & Monitoring:** Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring
- **Bug Analysis & Quality Intelligence:** Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting
- **Pipeline Integration:** Test automation in Azure DevOps CI/CD pipelines with HTML report upload to Azure Test Results

Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLAlchemy, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, Azure Blob Storage, Poetry

AI-driven Automated QA Environment for Energy Infrastructure

07/2024 - 01/2025

AkkuSwap Startup

QA Leadership for EU-wide Battery Swap Infrastructure: AI-driven simulation tool for battery swap station network with energy infrastructure integration

Key Responsibilities

- QA concept, design and implementation for AI-driven simulation of EU-wide infrastructure
- QA framework development for inhouse AI server infrastructure
- Proof-of-concept for automated testing of AI-driven simulations

Energy Sector Connection

- **50 Hertz Participation:** 50 Hertz (Elia Group subsidiary) participated in research project eHaul, the predecessor of AkkuSwap
- Direct energy sector experience with critical infrastructure quality requirements

Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Created testing framework for energy sector critical systems
- Validated proof-of-concept with measurable reliability improvements

Technical Lead & AI Automation Architect

01/2024 - 04/2025

BKS

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

Focus areas:

- **Workflow automation:** Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction:** Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration:** Connection to European e-invoicing network via Storecove Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration:** Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.
- **Hexagonal Architecture:** Domain-driven design with Ports & Adapters for maximum testability and interchangeability of infrastructure components.

Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Storecove API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

AI Engineering Lead & ML Solutions Architect

04/2023 - 04/2024

BKS on behalf of Ryze

Project management & technical lead: Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation."

Focus areas:

- **Development of ML models:** Design and training of specialized models for effective email classification, implemented via Azure Functions.
- **API integrations:** Seamless connection to Microsoft Outlook 365 via Microsoft Graph API and to SAP systems for customer data checks.
- **Process automation:** Automatic moving of emails based on classification and specific business rules, including complaint handling.
- **DevOps and monitoring:** Implementation of CI/CD pipelines and use of Kibana for monitoring.
- **Development of a Streamlit-based user interface:** Improved testability and visualization of process results.

Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021 - 05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

Tools

- IntelliJ / Visual Studio Code
- Java / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- CI/CD / GitHub - reusable workflows
- Helm / Containers
- JIRA / Confluence
- Grafana
- Behaviour-Driven Development
- Consumer-Driven Contracts (Pact.io)

DevOps Quality Lead & Dashboard Architect

01/2017 - 05/2021

DB Vertrieb

Test management, test coordination, test automation - agile (Kanban, Scrum, SAFe): Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

Tools

- IntelliJ
- Java
- Cucumber / Cypress
- JMeter / Gatling
- CI/CD / GitLab CI / Jenkins
- Helm / Docker
- JIRA / Confluence
- Kibana / Grafana / Instana / Graylog
- Behaviour Driven Development
- Consumer Driven Contracts (Spring Cloud Contract)
- Ionic (hybrid app)
- Selenium

Earlier Positions

09/2015 - 01/2017	Performance Engineering Lead & Test Architect — DB Systel	Load and performance tests and analysis - test manager, test designer, analyst
04/2015 - 09/2015	Senior iOS Developer & Mobile Architect — Telekom	Senior iOS Developer
05/2009 - 03/2015	Quality Assurance Lead & Test Automation Architect — Siemens	QA, test management, test automation, requirements engineering
03/2009 - 05/2009	Performance Test Engineer & Load Testing Specialist — ING-DIBA	Load and performance tester
01/2006 - 12/2008	Performance Test Engineer & Quality Consultant — British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank	Load and performance tester / tester

Workshops & Presentations

Guest Lecture: AI-Assisted Software Development in Practice
TU Darmstadt — Information Systems Group · Lab course "AI Startup: From Idea to Execution" — 04/2026

Workshop: Development with Generative Language Models
Developer Workshop - Agent-based Software Development — 06/2025

Presentation: Efficient Documentation through Automation
Enterprise Presentation - AI-powered Documentation Workflows — 04/2025

Applications of AI in Business Context
AI Workshop for Business Integration — 2023